GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Taal

The agent that sells what the forecast says you'll throw away.

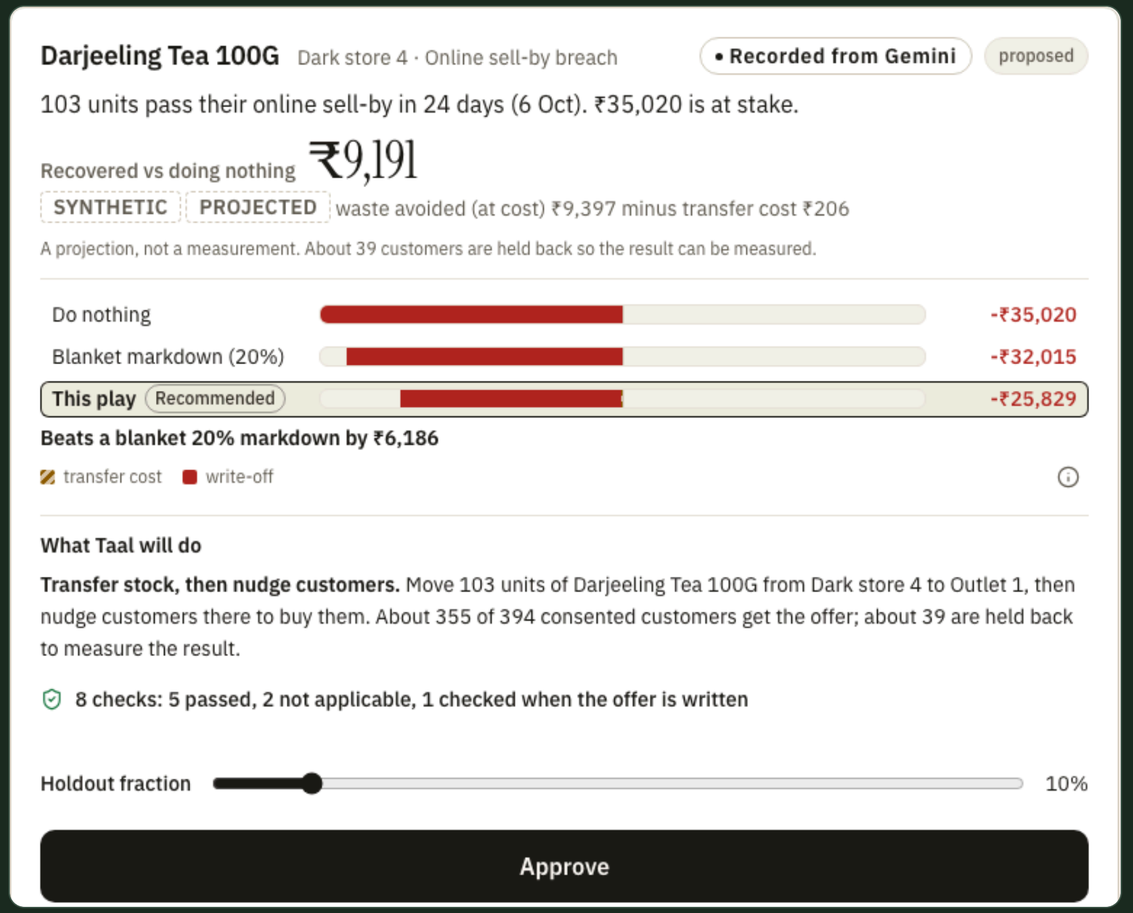
Taal spots stock at risk, plans a play with Gemini, a person approves, the offer goes out, and a holdout measures it. The agent is refused by code until a customer-targeted play carries a holdout; code assigns the groups, and the app says plainly when the result has not proved anything yet.

Context: Blinkit-scale inventory loss is about ₹0.5 cr per store a year (all causes). Taal projects ₹2.83 lakh per node a year at a 5% response rate (synthetic tenant).

₹9,191 projected recovery vs doing nothing on one tea lot; +₹6,186 vs a blanket markdown PROJECTED

For a dark-store manager clearing 103 packs of Darjeeling tea, 24 days from sell-by, ₹35,020 at stake. Synthetic data, labelled.

Nandish · lead **Rinu** · co-builder. We are in a category team at an Indian pet-care e-commerce company. The live demo runs on a synthetic tea tenant; the replay used our company's anonymised past data, with approval.



The live Play Desk, 16 Oct: the tea play, priced by code, waiting for a person to approve

REAL DATA · REPLAY · NOT A LIVE PILOT

Real data, 11 weeks: about ₹44.5 L pulled at expiry (INFERRED); Taal would have flagged 99% of it a month ahead (45% precision); estimated recovery ₹8 to 15 L, about twice a blanket markdown.

github.com/Nandish3010/probable-fortnight

THE PROBLEM · HOW BIG, WHICH RULE, WHO DECIDES

The forecast sees the write-off coming, but nothing acts on it.

HOW BIG · AS REPORTED

1.8%

of net order value (NOV) is lost to inventory at Blinkit.

As reported by Inc42, 22 Jul 2026. Per-store line below shows its inputs, from the same article.

₹17,132 cr ÷ 2,443 stores × 4 × 1.8% ≈ ₹0.5 cr per store a year

All inventory loss, one quarter annualised. Context for scale, not a Taal result.

THE RULE · AN ADVISORY, NOT LAW

30% or 45 days

of shelf life must remain at delivery of an online order.

FSSAI advisory to e-commerce food sellers, 3 Dec 2024. Stock can be in date on the shelf yet past its online sell-by, so the deadline can come weeks before expiry.

Taal plans to that online sell-by line, not the printed expiry. Either-limit reading is its configurable default.

WHO IT IS FOR

A dark-store manager with a sell-by lot

103 packs of Darjeeling tea, 24 days before the online sell-by, ₹35,020 at stake (seeded tenant).

A demand planner checks the guardrails and approves the play.

The category or supply-chain head owns write-off and buys Taal.

NEXT The loop and the hero result are on slide 4; projected impact is on slide 16; 3 of 7 practitioners we asked write off weekly or daily (slide 14).

REPLAY ON REAL DATA · A PET-CARE E-COMMERCE COMPANY · 21 JUL TO 19 OCT 2026, ONLINE

On real data: flagged a month ahead, in replay

Food and consumables, online. Replay of past data, not a live pilot.

DAYS BEFORE THE CUT-OFF · MEASURED

-30 d — **Taal would have flagged**
median 30 days before cut-off

-15 d — **Existing markdown trigger**
15 days before

0 — **Cut-off: stock pulled from sale**
owner's rule: remaining life below dispatch tolerance

A simple 60-days-of-stock rule fires about as early; Taal adds sizing the shortfall and choosing the action.

₹44.5 L
INFERRED, 11 weeks

About **₹44.5 L pulled at the cut-off in 11 weeks**, roughly ₹2 crore a year (INFERRED, annualised)

99%
would have been flagged; 45% precision

99% of those lots would have been flagged ahead, 87% at least a week ahead; 45% of in-window flags were pulled (MEASURED)

₹8 to 15 L
ESTIMATED

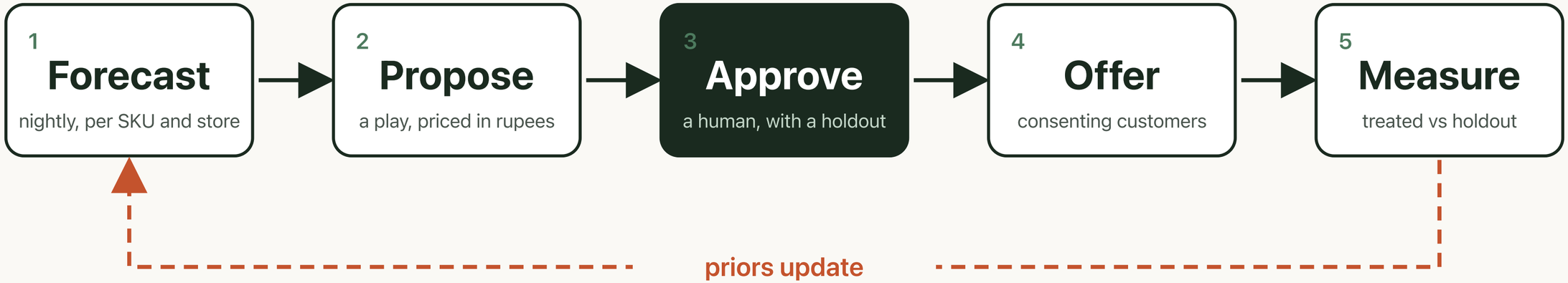
Transfer-or-markdown choice: estimated ₹8 to 15 L, about 2x a blanket markdown at the trigger (ESTIMATED)

LIMITS · Expiry loss inferred from stock leaving sellable inventory; cannot separate destruction, returns or miscounts.
Recall flattered: flag and label come from the same sales signal.
11 weeks; about 1,950 flagged lots reached the cut-off.
Values modelled from the company's own promo response; transfers at ₹40 a unit.

THE LOOP AND THE HERO PLAY

One worked example: a transfer plus targeted nudge recovers ₹9,191 vs doing nothing on one tea lot

PROJECTED · SEEDED TENANT



TEA · GEMINI RUN · PROJECTED

₹9,191

transfer plus targeted nudge (₹9,397 waste avoided – ₹206 transfer); ₹6,186 ahead of a blanket 20% markdown

HOLDOUT · SYNTHETIC ORDERS

+0.3 pp

INCONCLUSIVE, SYNTHETIC (pp = percentage points; 95% CI −8.7 to +1.6). A 5 pp lift needs about 1,156 customers (SIMULATION).

WHAT THIS PROVES

The loop closes

Measure runs end to end on synthetic orders; effect size needs a real pilot

THE HEADLINE

The first value is the right decision (transfer vs markdown); the holdout proves the offer part. Why tea: a generic dark-store tenant, so no employer data is used.

WHAT IS NEW · DIFFERENTIATION

No customer-targeted play without a holdout, enforced in code.

WHAT CLEARANCE TOOLS DO (PUBLIC VENDOR PAGES, READ 9 OCT)

- Price near-expiry stock down from its expiry date, in store
- Optimise markdowns and promotions inside enterprise planning suites
- Sell surplus food to consumers through a marketplace

WHAT TAAL ADDS

- Plans to the online sell-by line, not the printed expiry
- Every customer-targeted play carries a code-enforced holdout; price cuts use a node holdout when one exists and say so when it does not; an inconclusive result is reported as inconclusive and the measured rupee figure is withheld
- The approved play is written back into the forecast; each safety layer is measured separately

1 · Gemini proposes

mechanic, audience and rationale for the at-risk lot, in an ADK loop

2 · Code refuses it without a holdout

The rule (rule_holdout_required): reject if the holdout fraction is outside [0.05, 0.9], min_treated_n is unset, or the seed is missing; code then sets the seed and assigns the groups

3 · The result says what it proved

Tea: +0.3 pp, 95% CI −8.7 to +1.6, INCONCLUSIVE, SYNTHETIC. About 1,156 customers needed for a 5 pp lift (SIMULATION)

Each safety layer measured on its own: the tool refused every cross-customer call (the model tried 3 of 3 times); Model Armor blocked 3 of 8 chat injection cases and fails open; the cropped-label case (V6) fixed in code on 16 Oct. **Method:** protocol committed before any model call, a run is INVALID if a fallback model answers, spend cap enforced. **Why it compounds:** every measured play updates that retailer's response prior for its mechanic and category; a tool that skips the holdout has nothing to learn from.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.

Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

Customer Agent

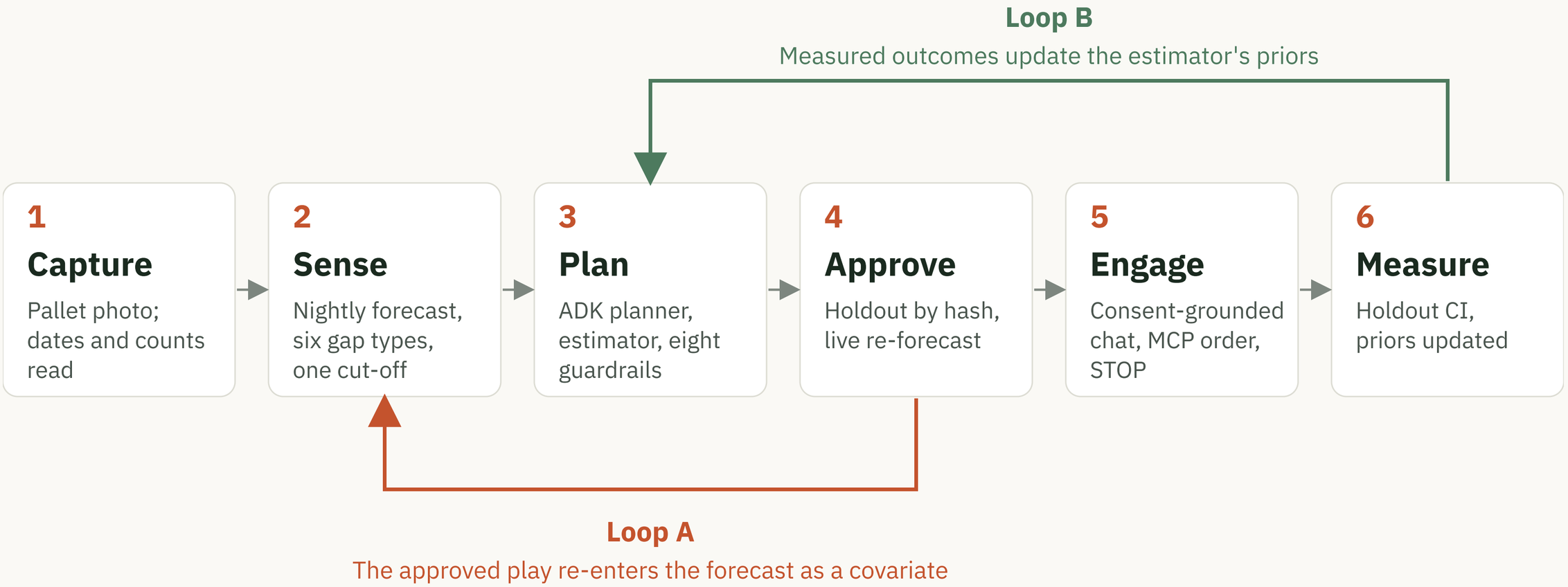
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

Measure

Treated vs holdout, 95% CI. Priors update from the counts.

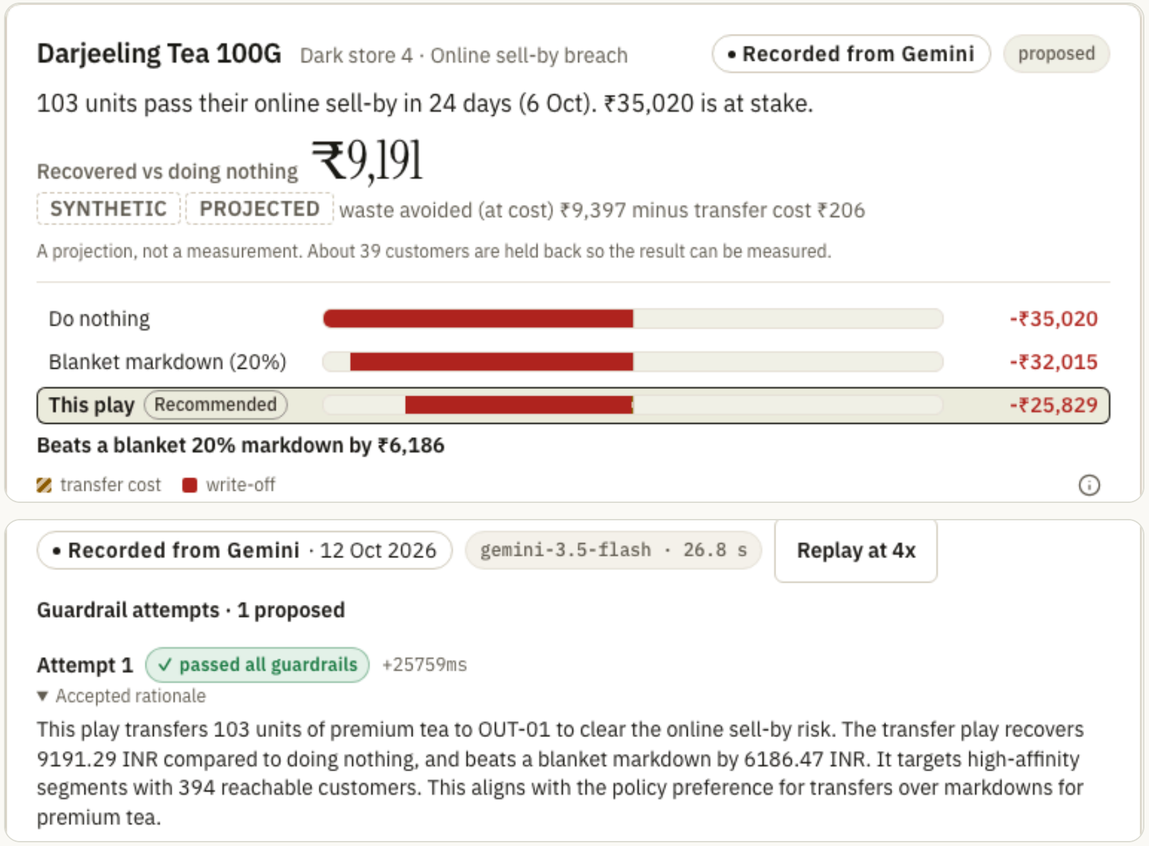
PROCESS FLOW

Two closed loops, not an open-ended pipeline.



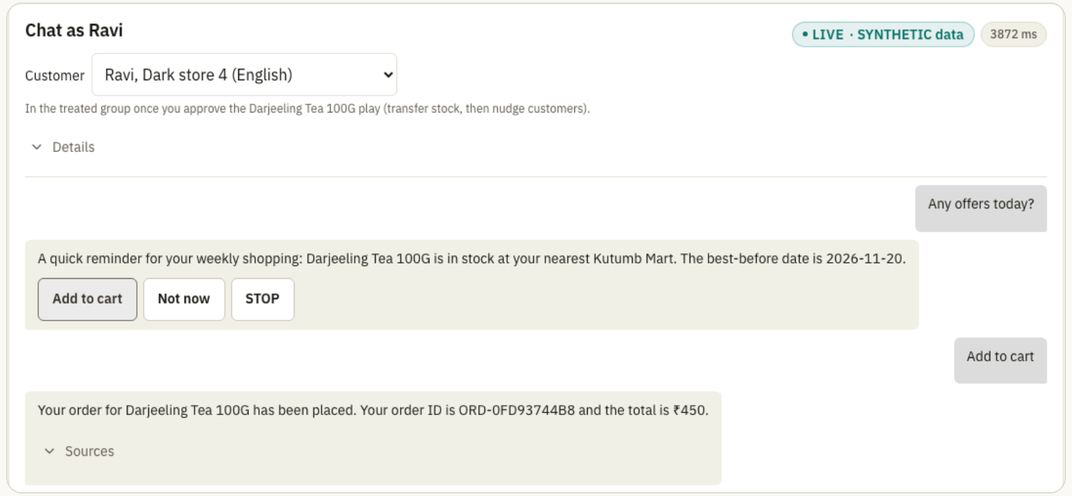
SCREENS

Gemini drafts the play; code prices every rupee; the customer orders in chat.



Play Desk, live app, 16 Oct

Gemini drafted the play and its rationale (recorded 12 Oct); code priced every rupee and the guardrails passed it on the first attempt. Projected, synthetic.

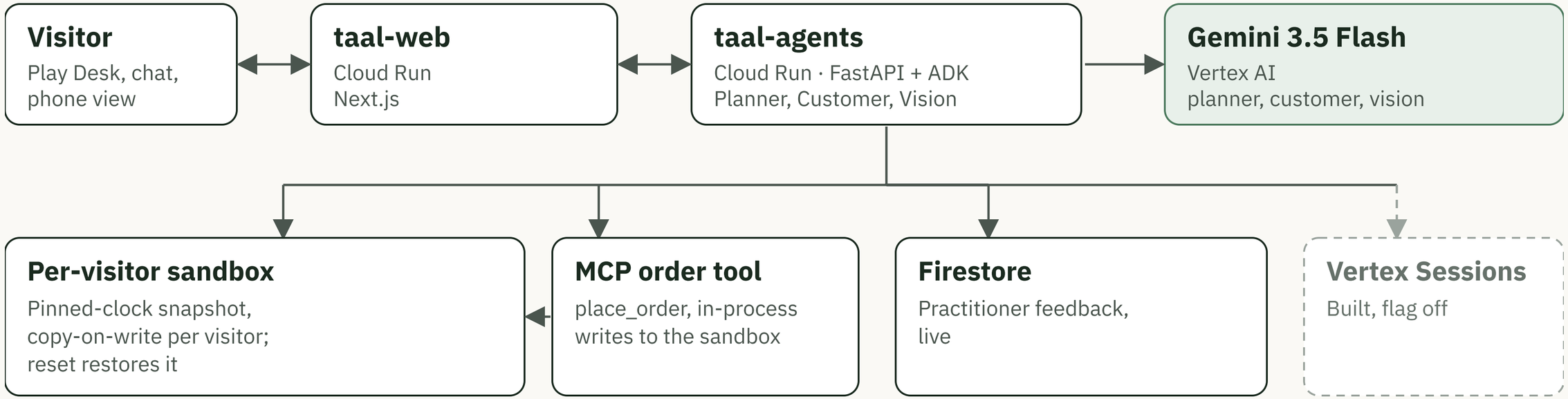


Chat as Ravi, live app, 16 Oct

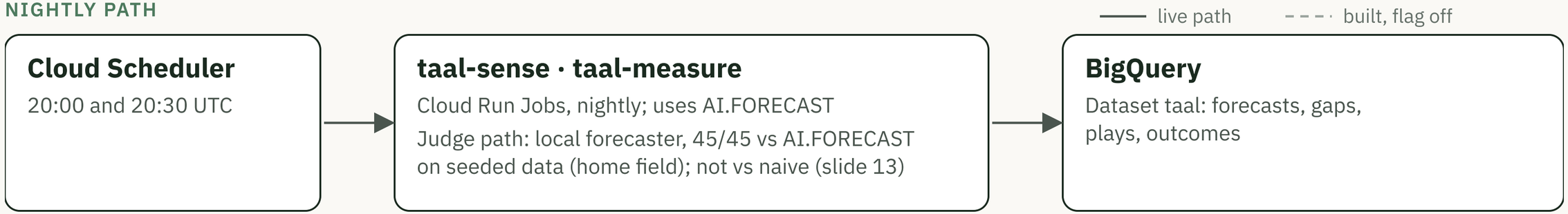
Ravi is in the treated group: the offer arrives, Add to cart places a ₹450 order through the MCP order tool. Replies also come in Kannada with an English gloss.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Each Google service, and the decision it makes.

SERVICE	THE DECISION IT MAKES
Google ADK planner (Gemini 3.5 Flash)	Proposes the play: mechanic, audience, window, in a LoopAgent (max 3) with 6 tools
Gemini on Vertex AI	Drafts the offer, chats with the customer, reads the pallet photo
Model Armor	Screens every chat message and reply for injection and jailbreak (live 14 Oct); fails open, code guardrails still apply
BigQuery AI.GENERATE_TABLE	Pre-generated the offer copy for the 7 seeded plays; stored and reused at Approve. Other plays try one call within 3 s, then templates.
Local forecaster (BigQuery AI.FORECAST nightly)	Re-forecasts the write-off on Approve; the nightly batch feeds Sense
Firestore	Stores practitioner feedback, live in the deployed service
Cloud Run, Jobs and Scheduler	Serves the app and API; runs Sense and Measure nightly; scales to zero when idle
MCP order tool	Places the customer's order when they accept the offer
Deterministic guardrails and estimator	Check every play against 8 rules and own every rupee; Gemini never sets a rupee figure

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PLANNER COST PER PLAY · 14 OCT

₹6.5 / $0.08

gemini-3.5-flash, token usage priced at list rates. Modelled, not a bill.

Offer copy is generated once and stored, so Approve on the seeded plays waits on no model call. Services scale to zero between visits. Production path: docs/scale.md.

₹486 / 30 days (23 Sep, pre-switch) was project spend, not per play.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.
Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

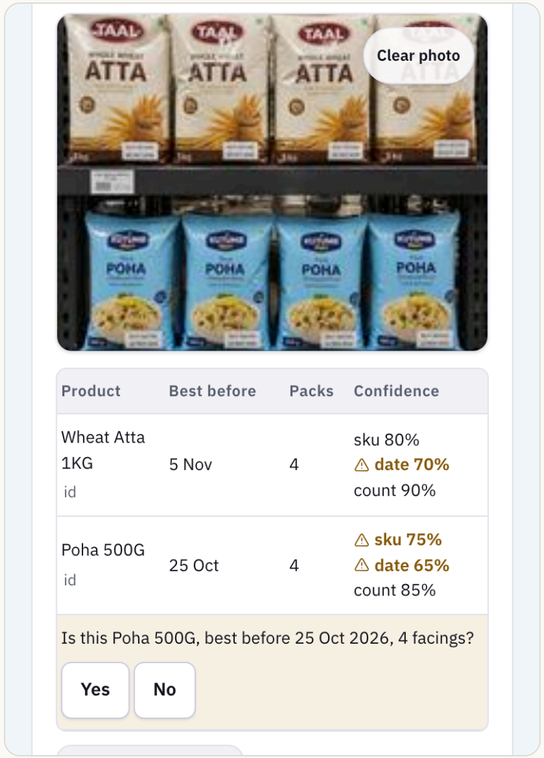
PROJECTED MARGIN LESS RUN COST ILLUSTRATION

≈ ₹21,710 – ₹156 to ₹313 per node

Per 28 days at the 5% prior: ₹347,356 margin over 16 nodes, less Mid-profile cost (₹2,500–5,000 a month, estimate). Synthetic.

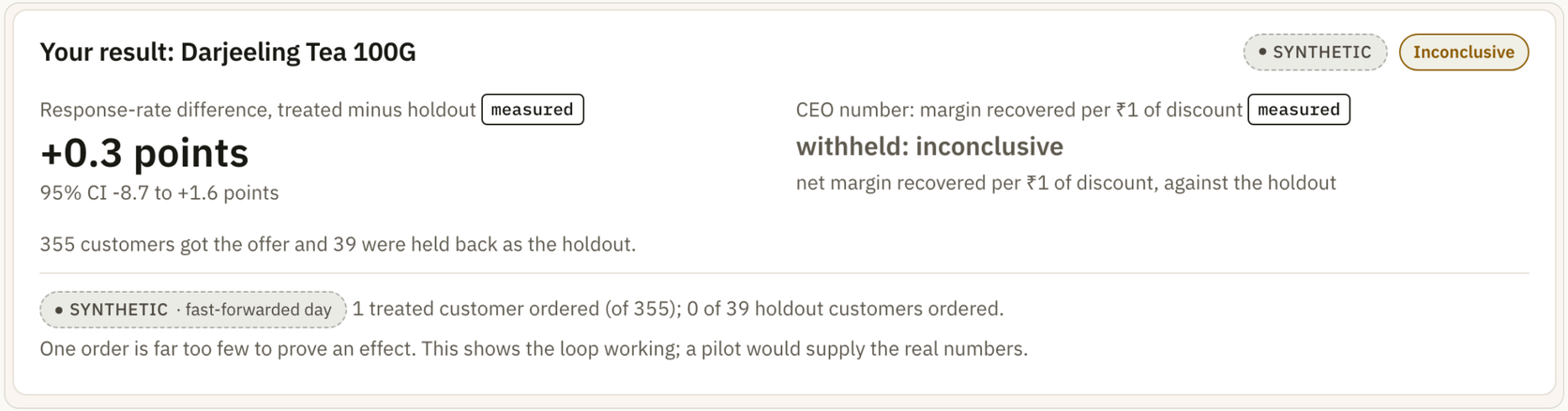
SNAPSHOTS OF THE PROTOTYPE

Doubtful reads go to a person; outcomes say SYNTHETIC.



Phone view

A read under 70% confidence is asked back before it reaches inventory.



Outcomes on a fast-forwarded day, synthetic orders: Tea

355 offered, 39 held out; 1 treated customer ordered, 0 held-out. Response rate +0.3 percentage points, 95% confidence interval –8.7 to +1.6. The range crosses zero, so the page says inconclusive and withholds the rupee figure.

PERFORMANCE REPORT

Measured and dated, on gemini-3.5-flash.

LIVE PLAN — 11 Oct

22/24

plans written by the model on the deployed app, 0 fallbacks; prompt v6, v7 ships

plan_live_2026-10-12.json

CUSTOMER CHAT — 14 Oct

3.66 s p50

p95 4.36 s, down from 8.44 s; Model Armor on

customer_latency_2026-10-14_before_after

APPROVE — 15 Oct

1.4–1.9 s

warm, recorded play, no model call; ten samples, 13 to 15 Oct; 3.3 s before

approve_timing_2026-10-15.json

PLANNER EVALSETS — 14 Oct

39/46

first proposal accepted by the guardrails

live_eval_2026-10-14/summary.json

ADVERSARIAL EVAL — 14 and 16 Oct

50 of 60 attack runs stopped

0 of 18 genuine requests refused. 10 of 60 runs (4 cases) got through; 3 of 4 fixed and re-run; the 4th (C8) refused by the tool every run.

eval/evaluation.md (adversarial; Hardening, 16 Oct)

PUBLIC BACKTEST, UCI ONLINE RETAIL II — 14 Oct

WAPE 1.12 vs 0.99 naive seasonal

The forecaster does not beat a naive seasonal baseline here. The loop after the forecast is the product.

eval/external_backtest.md · external_backtest_v2_2026-10-14.json

REAL-DATA REPLAY (19 OCT) — Replay, not a pilot

Would have flagged 99% of expiry pull-outs a median 30 days ahead; precision 45% (MEASURED). Replay of past data, not a pilot.

11 weeks, online food and consumables · eval/raw/shadow_pilot_2026-10-19/summary.md

Also measured: photo intake 30/30 dates read on synthetic photos, not real phone photos (14 Oct, vision_synthetic.json); 1,700+ tests (870 pytest + 833 Playwright, collected 16 Oct) in make verify before every deploy. Method and every source: eval/evaluation.md, deck/claims.md.

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

What seven practitioners told us.

3 of 7
write off stock weekly or daily

3 of 6
online sellers say the sell-by rule limits what they sell

0 of 7
measure a promotion against a control group

4 of 7
want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

Three people, three screens, one approved play.

Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

Badges. LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

Projected margin hinges on one assumption.

REAL DATA about ₹2 crore a year pulled at expiry in one company's online food and consumables (INFERRED, annualised from 11 weeks)

TWO LOTS · PROJECTED

Tea (recorded Gemini run)

₹9,191 recovered

Vs doing nothing; projected, seeded tenant. ₹6,186 ahead of a blanket 20% markdown. 103 units, ₹35,020 at stake.

Chips (10 Oct, API snapshot)

-₹279

Projected. 368 units, ₹9,200 at stake. Net −₹8,468 vs −₹8,189.

RESPONSE PRIOR SENSITIVITY · PROJECTED

PRIOR	MARGIN	PER NODE / YEAR
2%	₹1.41 lakh	₹1.15 lakh
5% default	**₹3.47 lakh**	**₹2.83 lakh**
10%	₹6.19 lakh	₹5.04 lakh

Seeded tenant, 28-day run, 550 gaps. Per node / year = margin ÷ 16 nodes × 365/28.

MEASURED VS PROJECTED

Measured on real customers: none yet.

Synthetic holdout, Tea: 355 treated, 39 held back; 1 responder vs 0, inconclusive.

Power (SIMULATION): a 5 pp lift needs about 1,156 customers at a 90/10 split.

Projected at 5%: ₹347,356 margin, ₹706,183 waste avoided.

Practitioners: 7 usable responses; 3 of 7 write off stock weekly or daily.
Scale of the problem: sourced loss figure and arithmetic on slide 2 (as reported, all inventory loss, not a Taal result).

LONGEVITY · RUNNING IT FOR REAL

What it takes to run beyond the demo.

Production path

Today: services run as the owner-role deploy identity; CI uses a stored key. Secrets sit in a managed store. Fair-use limits cap plan, chat and photo calls per visitor and in total.

Next: least-privilege service accounts and workload identity.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

How we kept it fast

Offer copy generated ahead and stored, so Approve takes 1.4 to 1.9 s warm (recorded play, no model call; 13 to 15 Oct; 3.3 s before).

Chat p95 8.4 s to 4.4 s (14 Oct): stock prefetch, client reuse, minimal thinking.

Services scale to zero; the first call after idle takes longer. Breaks first: max-instances 1 with re-plan state in memory.

Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Planner: about ₹6.5 per play at list prices (gemini-3.5-flash, 14 Oct); offer copy generated once and stored.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

CONNECTING TO A RETAILER The write-off owner (category or dark-store ops) pays. Stock with expiry and sales load from CSVs today; consented customers are a pilot's first step. Opted-in customers only, STOP in code, designed to India's DPDP consent model (not certified). Every play needs human approval.

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

Next: a real pilot, real photos, more tenants.

1 A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

2 Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 (14 Oct) is on synthetic ones.

HARNESS BUILT

3 Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LEARNING CURVE · MEASURED, CHANGED, RE-MEASURED

MEASURED	BEFORE	WHAT WE CHANGED	AFTER
Chat p95, same 24 prompts	8.44 s	stock prefetch, Model Armor client reuse, minimal thinking	**4.36 s, Model Armor on (14 Oct)**
Approve, warm	3.3 s (12 Oct)	offer copy generated ahead of Approve	**1.4–1.9 s (13 to 15 Oct, ten samples)**
Live planner	no_play, 110.9 s (20 Sep, pre-switch)	gap lookup moved into code	**22 of 24 model-authored (11 Oct)**
Planner evalsets	43/46 match, 1/46 trajectory (old recordings)	re-recorded on gemini-3.5-flash	**46/46, 18/46: regression guard, not accuracy**
Adversarial eval	10 of 60 runs (4 cases) got through	code-only fixes	**3 of 4 fixed and re-run; C8 refused by the tool every run**

Rows are copied from docs/learning_curve.md, which also lists the changes that made a number worse. Pre-switch means gemini-2.5-flash.

LINKS

Taal: the agent that sells what the forecast says you'll throw away

DEMO VIDEO

drive.google.com/file/d/1hBKapntHMnc4FNybRqSEWods2MTrluBw

GITHUB REPOSITORY

https://github.com/Nandish3010/probable-fortnight

LIVE APP

https://taal-web-2obkp776ca-el.a.run.app

Transfer plus targeted nudge: ₹9,191 projected recovery vs doing nothing on one tea lot · seeded tenant

22 of 24 live plans on gemini-3.5-flash, p50 26.5 s (11 Oct)

Holdout so far: synthetic and inconclusive; a real pilot is next

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.
Then, as Ravi, type: “Is the Darjeeling tea in stock?” · “Do you have Cola Zero?” · “STOP”

Team Taal Google Cloud AI Builder Cup 2026 · Retail & Commerce